Survival-supervised Bayesian matrix factorization for one or more data blocks: model, objective and updates

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This document is the specification behind the `ssbmf` R package. Sections 1–2 state the model and the choices it fixes; Section 3 states the objective and the treatment of the Cox term; Sections 4–6 derive every update and the start; Section 7 gives the algorithm. `verify_derivation.R` in this folder checks the bound of Section 3 and the stationarity of the updates numerically. Choices marked **provisional** were made without review and are listed in Section 9.

1 Generative model

Samples $i = 1, \dots, n$; blocks $m = 1, \dots, M$ with features $j = 1, \dots, p_m$; programs $k = 1, \dots, K$. Block m is the $n \times p_m$ matrix Y_m (samples by features internally; the package accepts either orientation).

$$Y_m = \mathbf{1}\mu_m^\top + LF_m^\top + E_m, \qquad (E_m)_{ij} \sim N(0, 1/\tau_{mj}),$$

$$L_{ik} \sim g_{L,k}, \qquad (F_m)_{jk} \sim g_{m,k}, \qquad \beta_k \sim g_\beta,$$

with all priors independent across entries given their parameters. Survival (t_i, δ_i), with an optional stratum s_i, follows a Cox model whose log partial likelihood (Breslow handling of ties) is

$$\ell_{\mathrm{PL}}(\eta) = \sum_{i:\delta_i=1} \Big[\eta_i - \log \sum_{j \in R_i} e^{\eta_j}\Big], \qquad R_i = \{j : t_j \geq t_i,\ s_j = s_i\},$$

with the projection predictor

$$\eta_i = \sum_k \beta_k Z_{ik}, \qquad Z_{ik} = \sum_m \sum_j \tilde{y}_{m,ij} (F_m)_{jk}, \qquad \tilde{Y}_m = Y_m - \mathbf{1}\bar{y}_m^\top.$$

$\bar{y}_m$ are the training feature means, fixed before fitting, so a new patient is scored as $\eta^{\mathrm{new}} = \sum_m (y_m^{\mathrm{new}} - \bar{y}_m)^\top F_m \beta$ without refitting. Survival depends on F and β, not on L.

The likelihood is raised to block weights w_m and a survival weight w_S:

$$p_w(Y, t, \delta \mid \cdot) \propto \prod_m p(Y_m \mid \cdot)^{w_m}\ \exp\{w_S\, \ell_{\mathrm{PL}}(\eta)\}.$$

The default is $w_m = w_S = 1$, the unweighted Bayesian model. The symbols are carried through every formula so that a balance rule (Part 4 of the rebuild plan) changes no derivation.

2 Choices fixed by the model

Feature intercept. Each block has its own μ_m, estimated. With an intercept, nonnegative or sparse loadings can model data on its natural scale (log expression, methylation β-values); the old single-modality model needed per-cohort z-scoring partly because it had none. Cohort differences in location are not modeled in μ (one intercept per feature); cohorts enter only as Cox strata (D4).

Prior families. $g_{L,k}$ is point-exponential (activities are nonnegative, a point mass at zero lets a program switch off). For F_m there are two candidates. Point-exponential gives nonnegative loadings: each program can only raise features above μ_m, so values below average must be explained by a lower intercept and other programs. Point-Laplace gives signed, sparse loadings: a program can raise some features and lower others, which matches centered data and methylation, at some cost in interpretability. **Provisional default: point-Laplace for** F_m, because it can represent

below-average values directly; both families are run in the simulation panel. g_β is a Normal $N(0, \sigma_\beta^2)$ shared by all programs, with σ_β^2 fitted by marginal likelihood (Section 4.4).

Identifiability and scale. The likelihood is unchanged by $(L_{\cdot k}, F_{\cdot k}, \beta_k) \to (L_{\cdot k}/c, cF_{\cdot k}, \beta_k/c)$ for $c > 0$, and the fitted priors are scale families, so the objective is nearly invariant along this direction. Nothing is rescaled inside the fitting loop (as in flashier): every rescaling would be a step outside the objective. Reported output uses a canonical scale: each program's projection score Z_k has unit standard deviation over the training samples, $F_{\cdot k}$ is multiplied and $L_{\cdot k}$ and β_k divided by $\mathrm{sd}(Z_k)$. This leaves $LF_m^\top$ and η unchanged, so `predict()` is unaffected.

Meaning of K. K is an upper bound. A point mass in $g_{L,k}$ or $g_{m,k}$ can switch a program off during fitting. After fitting, each program receives a role from posterior quantities: *survival-active* if the local false sign rate of β_k is at most 0.05; *reconstruction-only* if it is not survival-active and explains at least 0.1% of the centered variance of some block; *inactive* otherwise. Nothing is deleted or refitted.

Reduction to one block. With $M = 1$ the model is the single-modality model: $Y = \mathbf{1}\mu^\top + LF^\top + E$, $\eta = \tilde{Y}F\beta$. There is one implementation. It differs from the frozen benchmark configuration (YFB with per-cohort z-scoring, DeSurv gene selection, $K = 7$, $\alpha_F = 0$) in these ways: (i) a feature intercept is fitted; (ii) survival enters the update of F with weight w_S (the benchmark used $\alpha_F = 0$); (iii) no canonicalization or sign correction inside the loop; (iv) β has a shared Normal prior fitted by marginal likelihood and a Gaussian posterior; (v) the Cox term is treated by the bound of Section 3 instead of a working quadratic; (vi) loadings may be signed. The expression- only real-data comparison is therefore a non-inferiority check (O5), not a reproduction. The input matrix is whatever preprocessing produces; for the first real-data runs this is the benchmark's per-cohort z-scored matrix.

3 Objective

3.1 Variational family

Mean field: $q = \prod_{ik} q(L_{ik}) \prod_{m,j,k} q(F_{m,jk}) \prod_k q(\beta_k)$, with μ_m, τ_m and the prior parameters point-estimated. Write $\bar{\ell}_{ik} = EL_{ik}$, $\ell_{ik}^{(2)} = EL_{ik}^2$, $\bar{f}_{m,jk}$, $f_{m,jk}^{(2)}$ likewise, and $q(\beta_k) = N(b_k, s_k^2)$.

3.2 The bound

$$\mathcal{L}(q, \mu, \tau, g) = \sum_m w_m \mathcal{R}_m + w_S \mathcal{S} - \sum_k \mathrm{KL}_{L,k} - \sum_{m,k} \mathrm{KL}_{m,k} - \sum_k \mathrm{KL}_{\beta,k},$$

where each KL is $\mathrm{KL}(q \,\|\, g)$ against the prior currently held (the prior refitted at the last update of that column; Section 4.5).

Reconstruction. With residual $r_{m,ij} = y_{m,ij} - \mu_{mj} - \sum_k \bar{\ell}_{ik}\bar{f}_{m,jk}$,

$$\mathcal{R}_m = \sum_j \Big[\tfrac{n}{2} \log \frac{\tau_{mj}}{2\pi} - \tfrac{\tau_{mj}}{2} \sum_i \Big(r_{m,ij}^2 + \sum_k (\ell_{ik}^{(2)} f_{m,jk}^{(2)} - \bar{\ell}_{ik}^2 \bar{f}_{m,jk}^2) \Big) \Big].$$

This is exact: $E_q(y - \mu - \sum_k L_k F_k)^2$ under mean field.

Moments of the predictor. η_i is bilinear in F and β. Under mean field, with $\bar{Z}_{ik} = \sum_{m,j} \tilde{y}_{m,ij} \bar{f}_{m,jk}$ and $V_{ik} = \sum_{m,j} \tilde{y}^2_{m,ij} (f^{(2)}_{m,jk} - \bar{f}^2_{m,jk})$,

$$\bar{\eta}_i = \sum_k b_k \bar{Z}_{ik}, \qquad v_i = \mathrm{Var}_q\, \eta_i = \sum_k [s_k^2 \bar{Z}^2_{ik} + (b_k^2 + s_k^2) V_{ik}].$$

The cross-program terms cancel because programs and features are independent under q.

Cox term. $E_q \ell_{\mathrm{PL}} = \sum_{i:\delta_i=1} [\bar{\eta}_i - E_q \log \sum_{j \in R_i} e^{\eta_j}]$. The second expectation has no closed form. Three treatments were considered.

(a) *Working quadratic* (the current code): expand ℓ_{PL} to second order at the current $\bar{\eta}$. This is an approximation, not a bound. Refreshing the expansion point changes the objective, and an audit of the current code found the refreshed objective lower in 18 of 50 refreshes. Coordinate ascent on it has no guarantee. Rejected.

(b) *Fixed-curvature bound* (Böhning). The Hessian of $\mathrm{LSE}(\eta_R) = \log \sum_{j \in R} e^{\eta_j}$ is $\mathrm{diag}(\pi) - \pi\pi^\top \preceq \frac{1}{2} I$, so for any anchor ψ

$$\mathrm{LSE}(\eta_R) \le \mathrm{LSE}(\psi_R) + \pi(\psi)^\top (\eta_R - \psi_R) + \tfrac{1}{4} \|\eta_R - \psi_R\|^2.$$

Taking E_q gives an exact lower bound on $E_q \ell_{\mathrm{PL}}$ that is quadratic in every coordinate, maximized over ψ at $\psi = \bar{\eta}$. After that maximization the bound equals $\ell_{\mathrm{PL}}(\bar{\eta}) - \frac{1}{4} \sum_j d_j v_j$, with d_j the number of events whose risk set contains j. The exact second-order term is $\frac{1}{2} \sum_j P_j v_j$ with $P_j = \sum_{i:\delta_i=1, j \in R_i} \pi_{ij}$, of order the cumulative hazard (about 1), while $d_j/2$ is of order the number of events (tens to hundreds). The bound therefore inflates the precision that survival contributes to $q(\beta)$ and $q(F)$ by one to two orders of magnitude: posterior intervals for β would be far too narrow (O7) and the steps very short. Kept only as a reference in `verify_derivation.R`.

(c) *Moment (Jensen) bound.* By Jensen's inequality, $E_q \log \sum_j e^{\eta_j} \le \log \sum_j E_q e^{\eta_j}$, and if $q(\eta_j)$ is Gaussian, $E_q e^{\eta_j} = \exp(\bar{\eta}_j + v_j/2)$. This gives

$$\mathcal{S} = \sum_{i:\delta_i=1} \Big[\bar{\eta}_i - \log \sum_{j \in R_i} \exp\left(\bar{\eta}_j + \tfrac{1}{2} v_j\right) \Big].$$

Its second-order behaviour in v is $\frac{1}{2} \sum_j P_j v_j$, the correct order. Two caveats. First, $q(\eta_j)$ is exactly Gaussian only if F is held fixed; with random F it is a sum over many features of products, close to Gaussian when many features load, so (c) is a bound under a Gaussian approximation. Second, Jensen treats the η_j within a risk set as if they varied independently, while they share β; for β_k this replaces the within-risk-set variance of Z_k by its uncentered second moment, a small overstatement of the precision when Z_k is centered.

Provisional choice: (c). It has the correct curvature, so $q(\beta)$ has calibrated width (O7) and survival steps are not artificially short; its caveats are checked numerically (`verify_derivation.R` compares (b) and (c) with Monte Carlo values of $E_q \ell_{\mathrm{PL}}$ at realistic states, including point-Laplace F and ties). $\mathcal{L}$ with (c) is *the* objective: a deterministic function of the variational moments, evaluated exactly after every update. Monotonicity (O1) is stated and tested for $\mathcal{L}$.

4 Updates with the Cox term

4.1 Derivatives of $\mathcal{S}$

Let $a_j = \bar{\eta}_j + v_j/2$, $\pi_{ij} = e^{a_j} / \sum_{l \in R_i} e^{a_l}$ for $j \in R_i$, $P_j = \sum_{i:\delta_i=1,\ j \in R_i} \pi_{ij}$ and the residual $\rho_j = \delta_j - P_j$ (a martingale residual with the moment weights). Then $\partial \mathcal{S}/\partial \bar{\eta}_j = \rho_j$ and $\partial \mathcal{S}/\partial v_j = -P_j/2$. All of π, P, ρ are computed in $O(n \log n)$ by sorting times within strata and taking cumulative sums in log space.

4.2 Conjugate-computation form

$q(F_{m,jk})$ enters $\mathcal{S}$ only through its first two moments $(\bar{f}, f^{(2)})$. Holding everything else fixed,

$$\frac{\partial \mathcal{S}}{\partial f^{(2)}_{m,jk}} = -\tfrac{1}{2}(b_k^2 + s_k^2) \sum_i P_i \tilde{y}^2_{m,ij} \equiv -\tfrac{1}{2} A_{m,jk},$$

$$\frac{\partial \mathcal{S}}{\partial \bar{f}_{m,jk}} = b_k \sum_i \rho_i \tilde{y}_{m,ij} - s_k^2 \sum_i P_i \bar{Z}_{ik} \tilde{y}_{m,ij} + A_{m,jk}\, \bar{f}_{m,jk} \equiv c_{m,jk}.$$

Replacing $\mathcal{S}$ by its first-order expansion in the moments, $c\,\bar{f} - \frac{1}{2} A\, f^{(2)} = E_q[cF - \frac{1}{2} AF^2]$, is a Gaussian pseudo-likelihood in $F_{m,jk}$ with precision A and linear coefficient c. This is conjugate-computation variational inference (Khan and Lin, 2017): the Gaussian "message" from the non-conjugate term is its gradient in the mean parameters. Its fixed points are stationary points of $\mathcal{L}$, and a step of size $\rho \in (0, 1]$ in the message, $(c, A) \leftarrow (1-\rho)(c, A)_{\text{old}} + \rho (c, A)_{\text{new}}$, is a natural-gradient step.

Safeguard. A full step need not increase $\mathcal{L}$. After every update of $q(F_{m,\cdot k})$ or $q(\beta)$ the objective is evaluated exactly. If it decreased, ρ is halved (up to 8 times); if no ρ increases $\mathcal{L}$, the update is rejected and q is left unchanged. Every accepted update therefore increases $\mathcal{L}$ by construction, and the fit records the number of halvings and rejections. Exact closed-form coordinate maxima (intercept, precisions, L) need no safeguard.

4.3 Update of $q(F_{m,\cdot k})$ and $g_{m,k}$

The reconstruction part is exactly Gaussian in $F_{m,jk}$, with precision and linear coefficient

$$a^{\mathrm{R}}_{m,jk} = w_m \tau_{mj} \sum_i \ell^{(2)}_{ik}, \qquad h^{\mathrm{R}}_{m,jk} = w_m \tau_{mj} \sum_i \bar{\ell}_{ik} \Big(y_{m,ij} - \mu_{mj} - \sum_{k' \neq k} \bar{\ell}_{ik'} \bar{f}_{m,jk'} \Big).$$

Adding the survival message gives the pseudo-observation

$$x_j = \frac{h^{\mathrm{R}}_{m,jk} + w_S\, c_{m,jk}}{a^{\mathrm{R}}_{m,jk} + w_S A_{m,jk}}, \qquad s_j^2 = \frac{1}{a^{\mathrm{R}}_{m,jk} + w_S A_{m,jk}}.$$

`ebnm` with the block's prior family, applied to $(x_j, s_j)_{j=1}^{p_m}$, returns the prior $g_{m,k}$ maximizing the marginal likelihood and the exact posterior under it; this jointly maximizes the surrogate over $(q(F_{m,\cdot k}), g_{m,k})$. The KL is recovered from the EBMF identity

$$\mathrm{KL}_{m,k} = \sum_j E_q \log N(x_j; F_{m,jk}, s_j^2) - \log p_{\hat{g}}(x),$$

with $\log p_{\hat{g}}(x)$ the `ebnm` log marginal likelihood. All features of the column use messages computed at the start of the update (a Jacobi step), which can overshoot when many correlated features move together; the safeguard handles this. With $w_S = 0$ the update is the unsupervised EBMF (flashier) update.

4.4 Update of $q(\beta)$ and σ_β^2

For $q(\beta_k) = N(b_k, s_k^2)$, mean parameters $(b_k, b_k^2 + s_k^2)$:

$$A_k^\beta = \sum_i P_i(\bar{Z}_{ik}^2 + V_{ik}), \qquad c_k^\beta = \sum_i \rho_i \bar{Z}_{ik} + b_k \sum_i P_i \bar{Z}_{ik}^2.$$

Pseudo-observations $x_k = c_k^\beta / A_k^\beta$, $s_k^2 = 1/(w_S A_k^\beta)$ for all k at once; `ebnm` with the Normal family (mode 0) fits σ_β^2 by marginal likelihood and returns $q(\beta_k) = N(\sigma_\beta^2 x_k/(\sigma_\beta^2 + s_k^2),\ \sigma_\beta^2 s_k^2/(\sigma_\beta^2 + s_k^2))$. The safeguard applies. The β update is iterated (up to 25 inner steps) until the messages stop changing, so each outer β update is close to a coordinate maximum.

Absorbing zero. A per-program Normal prior with variance set from the posterior alone (EM: $\sigma_k^2 = E\beta_k^2$) is absorbing: once $\sigma_k^2 = 0$, $q(\beta_k) = \delta_0$ and the next EM step returns 0. A shared point-Laplace prior whose slab weight $\hat{\pi}$ is fitted to near 0 has the same behaviour for all programs at once. Here σ_β^2 is fitted by *marginal* likelihood of the pseudo-observations, $\hat{\sigma}_\beta^2 = \arg\max \sum_k \log N(x_k; 0, \sigma^2 + s_k^2)$, jointly with q. The pseudo-observation x_k is computed from the data (the score $\sum_i \rho_i \bar{Z}_{ik}$ at the current predictor), not from the previous posterior, so $\hat{\sigma}_\beta^2 > 0$ as soon as one program's score exceeds its noise level, $x_k^2 > s_k^2$, whatever the current $q(\beta)$. Zero is therefore not absorbing: $\beta = 0$ is a fixed point only when no program carries survival signal at the current F. One gating effect remains: the directional survival message to $F_{\cdot k}$ is proportional to b_k, so a program with $b_k = 0$ receives no pull toward the survival direction. The survival-score initialization (Section 6) addresses this.

4.5 Prior refits and KL bookkeeping

Every prior refit happens inside an `ebnm` call that also returns the exact posterior under the refitted prior, and the stored KL for that column is computed against that prior. No KL is ever evaluated against a stale prior, and no prior is refitted without its posterior (the old multimodal code did both).

5 Updates without the Cox term

Intercept. $\mu_{mj} = \frac{1}{n} \sum_i (y_{m,ij} - \sum_k \bar{\ell}_{ik} \bar{f}_{m,jk})$, the exact maximizer.

Precisions. $\tau_{mj} = n / \sum_i E_q(y_{m,ij} - \mu_{mj} - \sum_k L_{ik} F_{m,jk})^2$, capped at $\tau_{\max,mj} = 10^6 / \widehat{\mathrm{Var}}(y_{m,\cdot j})$. $\mathcal{R}_m$ is concave in τ_{mj}, so the capped value is the constrained maximizer.

Activities. Survival does not involve L. For column k, pooling the blocks,

$$a^L_{ik} = \sum_m w_m \sum_j \tau_{mj} f^{(2)}_{m,jk}, \qquad h^L_{ik} = \sum_m w_m \sum_j \tau_{mj} \bar{f}_{m,jk} \Big(y_{m,ij} - \mu_{mj} - \sum_{k' \neq k} \bar{\ell}_{ik'} \bar{f}_{m,jk'} \Big),$$

and `ebnm` (point-exponential) on $x_i = h^L_{ik}/a^L_{ik}$, $s_i = (a^L_{ik})^{-1/2}$ gives the exact coordinate maximum over $(q(L_{\cdot k}), g_{L,k})$.

6 Initialization

1. Center each block at its training means and divide it by its root mean square, so blocks of different size and scale contribute comparably; concatenate the blocks and take the rank-K truncated SVD $UDV^\top$.
2. For nonnegative L: for each component take the sign whose positive part of $U_{\cdot k}$ has the larger norm, $L_{\cdot k} = \max(0, \pm U_{\cdot k})\sqrt{D_k}$ plus 10^{-3}, and set F_m to the least-squares loadings given L (clipped at 0 for point-exponential F).
3. $\mu_m = \bar{y}_m$, τ_{mj} from the initial residuals, posterior second moments equal to squared means, $q(\beta) = N(0, 0)$ and σ^2_β fitted at the first β update.
4. Optional (`init = "survival"`): replace the last program's loadings by the survival score direction, $F_{m,\cdot K} \propto \tilde{Y}_m^\top u^0$, with u^0 the null-model Cox martingale residuals, restricted to the sign allowed by the prior. Off by default.

7 Algorithm

```
input: blocks Y_m, (t,  , strata), K, weights w_m, w_S, tolerances
1  initialize (Section 6); compute
2  repeat for sweep = 1, 2, ..., max_sweeps:
3      _m,  _m for every block                              (exact)
4     for k = 1..K: q(L_·k), g_L,k                          (exact, ebnm)
5     for m = 1..M, k = 1..K: q(F_m,·k), g_m,k              (message step + safeguard)
6     q( ),  ²_ , iterated to message convergence           (message step + safeguard)
7     evaluate  ; record the change after every step of 3-6
8     stop when | _new -  _old| < tol · | _new|            (tol = 1e-6 by default)
9  report: moments, priors,  trace and its components, the number of
   steps that lowered  by more than 1e-8·| | (must be 0), halvings,
   rejections, convergence flag
```

The per-step check in line 7 is how O1 is measured.

8 Balance between survival and reconstruction

Under the Gaussian model each of the np_m entries of block m is an independent observation, while the Cox term carries information of the order of the number of events. The survival share of the precision of $q(F_{m,jk})$, $w_S A_{m,jk}/(a^{\mathrm{R}}_{m,jk} + w_S A_{m,jk})$, measures this directly (the fit reports its median over features, `surv_share`). Two rules are implemented, in the order of the rebuild plan; neither is used by default.

Effective number of features (`weight_rule = "effective_features"`, no tuning parameter). If features are correlated beyond the K programs, counting every entry as independent overstates the information in the block. With residuals R_m of the initial rank-K fit and $S = R_m^\top R_m / n_{\mathrm{df}}$ ($n_{\mathrm{df}} = n - K - 1$), the participation ratio $p_{\mathrm{eff}} = (\operatorname{tr} S)^2 / \operatorname{tr} S^2$ equals p_m for independent features of equal variance. Because $E \operatorname{tr} S^2 \approx \operatorname{tr} \Sigma^2 + (\operatorname{tr} \Sigma)^2 / n_{\mathrm{df}}$, the estimate uses $\operatorname{tr} S^2 - (\operatorname{tr} S)^2 / n_{\mathrm{df}}$, computed through the $n \times n$ Gram matrix. The block weight is $w_m = \min(1, p_{\mathrm{eff}}/p_m)$, fixed before the main loop, so the objective (with these weights) is unchanged in form and every monotonicity argument above holds.

Survival weight chosen by training cross-validation (last resort). `ssbmf_cv(w_grid = ...)` cross-validates w_S jointly with K; among settings within one standard error of the best mean C, the smallest w_S and then the smallest K are chosen. A power likelihood $\exp\{w_S \ell_{\mathrm{PL}}\}$ cannot in general be learned by maximizing the marginal likelihood (the weight is not a parameter of a proper model); calibration approaches for such weights (general Bayesian updating, Bissiri, Holmes and Walker; the choice of the loss scale, Holmes and Walker; SafeBayes, Grünwald) have not been checked against their sources here and are not implemented.

9 Provisional choices made without review

1. Cox treatment (c), the Gaussian-moment Jensen bound, with message (natural-gradient) updates and an exact-objective safeguard.
2. Point-Laplace default for F_m; point-exponential for L; shared Normal prior for β fitted by marginal likelihood.
3. Role thresholds: lfsr ≤ 0.05 for survival-active; 0.1% of a block's centered variance for reconstruction-only.
4. Precision cap $\tau \leq 10^6 / \widehat{\mathrm{Var}}(y_j)$; convergence tolerance 10^{-6} relative, 1,000 sweeps.
5. Initialization by scaled concatenated SVD; survival-score initialization off by default.
6. Canonical reporting scale: unit standard deviation of each program score.
7. Balance rules (Section 8) implemented but off by default; the effective- feature estimate uses initial residuals and the bias-corrected participation ratio.